

1-18 A new 1.5-V AA battery delivers 40 kJ of energy during its lifetime. How long will the battery last in an application that draws 10 mA continuously. Assume the battery voltage is constant.

1-23 Figure P1-22 shows an electric circuit with a voltage and a current variable assigned to each of the six devices. Use power balance to find v_4 when $v_1 = 20$ V, $i_1 = 2$ A, $p_2 = 20$ W, $p_3 = 10$ W, $i_4 = 1$ A, and $p_5 = p_6 = 2.5$ W. Is device 4 absorbing or delivering power?

1-28 Charge Storage Device

A capacitor is a two-terminal device that can store electric charge. In a linear capacitor the amount of charge stored is proportional to the voltage across the device. For a particular device the proportionality is $q(t) = 10^{-7}v(t)$. If $v(t) = 0$ for $t < 0$ and $v(t) = 10(1 - e^{-5000t})$ for $t \geq 0$, find the energy stored in the device at $t = 200$ μ s.

2-5 In Figure P2-5 find R_x and the power delivered to the resistor.

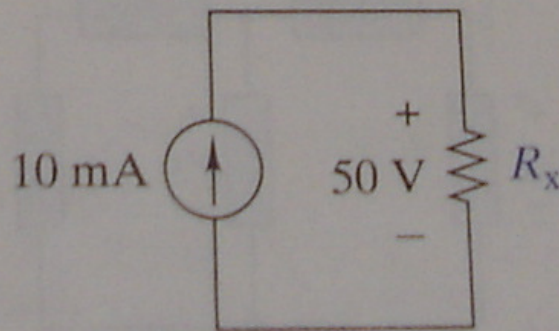


FIGURE P2-5

2-8 A certain type of film resistor is available with resistance values between $10\ \Omega$ and $100\ \text{M}\Omega$. The maximum ratings for all resistors of this type are 500 V and $\frac{1}{4}\text{ W}$. Show that the voltage rating is the controlling limit for $R > 1\ \text{M}\Omega$, and that the power rating is the controlling limit when $R < 1\ \text{M}\Omega$.

2-10 In Figure P2-10 $i_2 = 2 \text{ A}$ and $i_3 = -5 \text{ A}$. Find i_1 and i_4 .

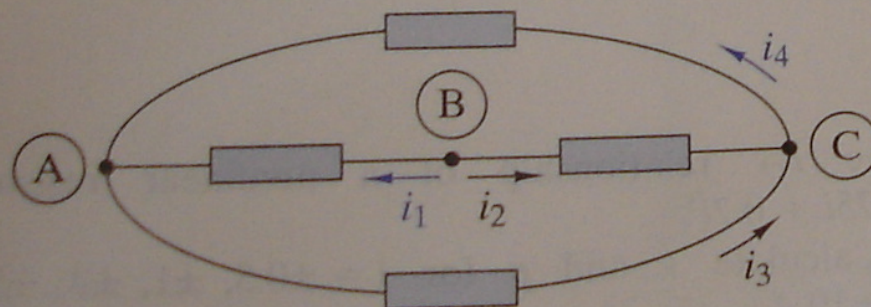


FIGURE P2-10

2-16 In Figure P2-16 $v_2 = 10 \text{ V}$, $v_3 = 10 \text{ V}$, and $v_4 = 10 \text{ V}$. Find v_1 and v_5 .

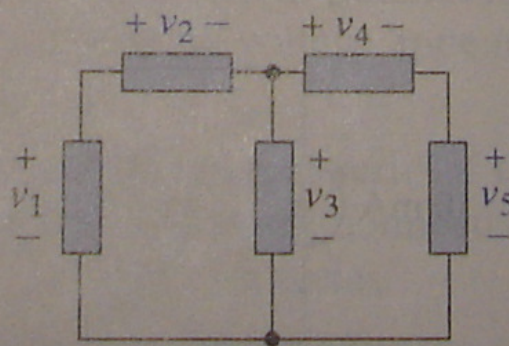


FIGURE P2-16

2-23 Find v_x in Figure P2-23.

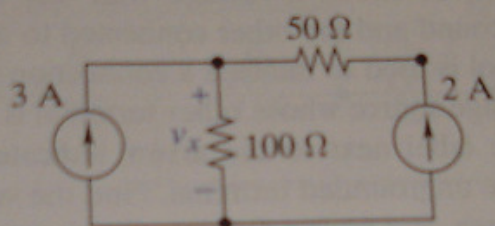


FIGURE P2-23

2-28 Find the equivalent resistance R_{EQ} in Figure P2-28.

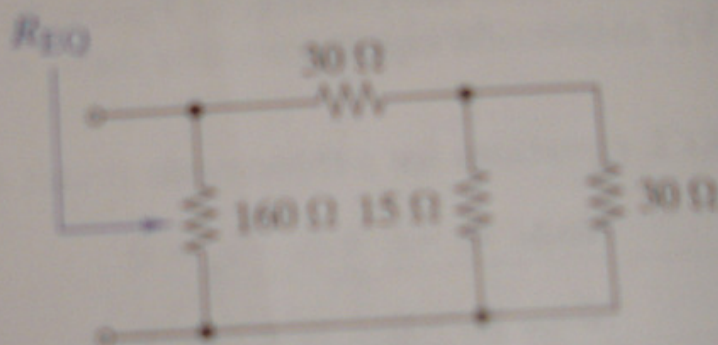


FIGURE P2-28

2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60\ \text{k}\Omega$.

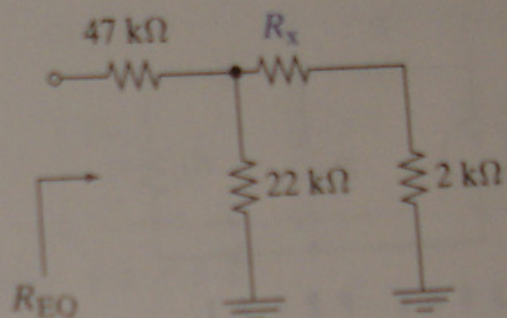


FIGURE P2-36